# Cart lab: change mass

**antocciphysics**

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Use a controlled comparison to describe the effect of system mass on acceleration.

**SUCCESS CRITERIA - I can...**

1. Identify total moving mass and keep driving force fixed.
2. Compare acceleration values using a fair test.
3. Separate model data from measured evidence.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Bring back an earlier idea

A car starts at +24 m/s and brakes at -4 m/s² until rest. A student extends that acceleration to 8 s. Find the stop time/distance and explain why extension is invalid for ordinary braking.

**Marzano check - circle:** ① ② ③

## 02 Explain a worked example

**Reference:** $a = F_{net}/M = m_h\, g/M$.
Ideal cart + hanger: M is all moving mass; $m_h$ is hanging mass. Hold $m_h$ fixed; neglect resistance and pulley inertia.

**Worked reasoning:** Supplied cart trials use the same net force: masses 1 and 2 kg accelerate at 4 and 2 m/s². In both cases ma=4 N; doubling mass halves acceleration.

**Your turn:** Plot acceleration against mass for the two trials. Label what was held constant.

**Marzano check - circle:** ① ② ③

## 03 Test an inverse relationship

The ideal data use the same 0.49 N net force. Compute 1/M for each row and describe the graph of a versus 1/M. Give slope units and physical meaning.

| M (kg) | a (m/s²) |
|---|---|
| 0.50 | 0.980 |
| 0.75 | 0.653 |
| 1.00 | 0.490 |
| 1.50 | 0.327 |

**Marzano check - circle:** ① ② ③

## 04 System boundary

Explain why the hanger belongs in total moving mass but string tension is internal to the cart-plus-hanger system. State two idealizations.

**Marzano check - circle:** ① ② ③

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### 05 Measured slope and repetitions

One supplied v-t interval goes from (0.20 s, 0.10 m/s) to(0.80 s, 0.40 m/s). Find a. Design a three-mass, three-trial data table for the actual lab; label any measurements you collect.

**Marzano check - circle:** 1 2 3

### 06 Choose a method

A 3 kg object hangs still from a vertical string. Use g=9.8 N/kg. Name all forces on it with sources, sizes and directions. Does object-on-string belong on this diagram?

**Marzano check - circle:** 1 2 3

### 07 Exit check

A 0.80 kg moving system accelerates at 0.60 m/s². At the same net force, predict a for 1.20 kg. Explain how added resistance would change the prediction.

**Marzano check - circle:** 1 2 3

### 08 STRETCH: Apply it another way

A cart model predicts a=F/m. At constant net force 3 N, find accelerations for masses 0.5, 1, 2 kg. Choose a transformed horizontal axis that makes a straight-line graph and state its slope.

**Marzano check - circle:** 1 2 3